

Introduction

For a time-to-event random variable $T \geq 0$, the **survival function** $S(t) = P(T > t)$ is the probability of not having experienced the event by time t . It is the complement of the CDF and the single most important descriptive quantity in survival analysis.

Prerequisites

Random variables, CDF, basic survival-analysis concepts.

Theory

For a non-negative random variable T ,

$$S(t) = P(T > t) = 1 - F(t).$$

Properties:

- $S(0) = 1$, $\lim_{t \rightarrow \infty} S(t) = 0$.
- Non-increasing.
- Right-continuous (by convention).

Related quantities:

- PDF: $f(t) = -S'(t)$.
- Hazard: $h(t) = f(t)/S(t)$.
- Cumulative hazard: $H(t) = -\log S(t)$.

Classical forms:

- Exponential: $S(t) = e^{-\lambda t}$.
- Weibull: $S(t) = \exp[-(t/\lambda)^k]$.
- Log-normal: $S(t) = 1 - \Phi((\log t - \mu)/\sigma)$.

Kaplan-Meier is the non-parametric MLE of S from right-censored data: a step function that drops at each observed event time by a factor reflecting the fraction of the at-risk set that experiences the event.

Assumptions

Non-negative random variable representing a time to an event. For Kaplan-Meier: independent censoring.

R Implementation

```
library(survival)
set.seed(2026)

# Theoretical survival functions
t_grid <- seq(0, 10, length.out = 400)
S_exp <- exp(-0.3 * t_grid)
S_weibull <- exp(-(t_grid / 3)^2)
S_lnorm <- 1 - plnorm(t_grid, meanlog = log(3), sdlog = 0.5)

plot(t_grid, S_exp, type = "l", col = "#6A4C93", lwd = 2,
     main = "Theoretical survival functions", xlab = "t", ylab = "S(t)")
lines(t_grid, S_weibull, col = "#2A9D8F", lwd = 2)
lines(t_grid, S_lnorm, col = "#F4A261", lwd = 2)
legend("topright", c("exponential", "Weibull k=2", "log-normal"),
```

```

col = c("#6A4C93", "#2A9D8F", "#F4A261"), lwd = 2)

# Empirical Kaplan-Meier from censored data
n <- 80
T_true <- rweibull(n, shape = 2, scale = 3)
C <- runif(n, 0, 10)
T_obs <- pmin(T_true, C)
delta <- as.integer(T_true <= C)

km <- survfit(Surv(T_obs, delta) ~ 1)
plot(km, conf.int = TRUE, main = "Kaplan-Meier estimate",
     xlab = "t", ylab = "S(t)", col = "#2A9D8F")
lines(t_grid, exp(-(t_grid / 3)^2), col = "#F4A261", lwd = 2)

# Median survival
summary(km)$table["median"]
qweibull(0.5, shape = 2, scale = 3)

```

Output & Results

```

median
2.499

[1] 2.498

```

Empirical and theoretical median survival times agree. The Kaplan-Meier step function closely tracks the true Weibull survival curve.

Interpretation

Reporting a median survival with its 95 % CI from a Kaplan-Meier estimate is the standard headline statistic of survival studies. Survival functions are typically compared across groups via log-rank tests and visualised together on the same axes.

Practical Tips

- Report medians with CIs when survival drops below 0.5; report specific-time survival (1-year, 5-year) when the study ends before the median is reached.
- Kaplan-Meier is distribution-free; parametric survival models (Weibull, log-normal) give smooth curves but commit to a shape.
- Include a risk table below Kaplan-Meier plots (`survminer::ggsurvplot(..., risk.table = TRUE)`).
- Under heavy censoring, the tail of the survival curve is poorly estimated; avoid over-interpreting late-follow-up differences.
- Survival functions never take values above 1 or below 0; if your empirical estimator returns such values, there is a bug.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots